

CBSE Class 12
Computer Science
Previous Year Question Paper 2016

Series: ONS

Set- 4

Code no. 91

- Please check that this question paper contains **16** printed pages.
- Code number given on the right hand side of the question paper should be written on the title page of the answer-book by the candidate.
- Please check that this question paper contains **7** questions.
- **Please write down the Serial Number of the question before attempting it.**
- 15 minutes time has been allotted to read this question paper. The paper will be distributed at 10.15 a.m. From 10.15 a.m. to 10.30 a.m., student will read the question paper only and will not write any answer on the answer script during this period.

COMPUTER SCIENCE

Time Allowed: **3** hours

Maximum Marks: **70**

General Instruction:

- Programming Language in Section A: C++
- Programming Language in section B: Python
- Answer either SECTION A or B, and SECTION C is compulsory.
- It is compulsory to mention on the page 1 in answer book whether you are attempting SECTION A or SECTION B.
- All** questions are compulsory within each section.

SECTION - A

1. (a) Out of following, find those identifiers, which cannot be used for naming Variables, Constants or Functions in a C++ program: 2 Marks

Total*Tax, double, case, My name, New, switch, Column 31, _ Amount.

Ans: Total*Tax -because of* symbol
 Double -because it is a keyword
 My Name -because of space
 Switch -because it a keyword

(b) Ronica Jose has started learning C++ and has typed the following program. When she compiled the following code written by her, she discovered that she needs to include some header files to successfully compile and execute it. Write the names of those header files, which are required to be included in the code. 1 Mark

```
void main ()
{
double X, times, Result;
cin>>X>>Times;
Result = pow (X, Times);
Cout<<Result<<endl;
}
```

Ans: iostream.h
 Math.h

(c) Rewrite the following C++ code after removing any/all syntactical errors with each correction underlined. 2 Marks

Note: Assume all required header files are already being included in the program.

#define Formula (a,b) = 2*a + b

```
void main()
{
float X = 3.2; Y = 4.1;
Z = Formula (X, Y);
Cout<< 'Result='<<Z<<endl;
```

```
}
```

Ans: #define Formulas(a, b)=2*a+b

```
void main
{
    Float X=3.2, Y=4.1,Z;
    Z=Formula(X, Y);
    Cout<<"Results="<<Z<<endl;
}
```

(d) Find and write the output of the following C++ program code: 2 Marks

Note: Assume all required header files are already included in the program.

```
typedef char TEXT [80];
```

```
void jumbleUp (TEXT T)
{
    int L = strlen (T);
    for (int C=0; C<L-1; C+=2)
    {
        char CT = T [C];
        T [C] = T [C+1];
        T [C+1] = CT;
    }
    for (C=1; C<L; C+=2)
    if (T[C]>= 'M' && T[C]<= 'U')
        T [C] = '@';
}
```

```
void main()
{
    TEXT Str = "HARMONIOUS";
    JumbleUp (Str);
    cout<<Str<<endl;
```

```
}
```

Ans: AHM@N@OS@

(e) Find and Write the output of the following C++ program code: 3 Marks

Note: Assume all required header files are already being included in the program,
class Share

```
{
    long int code;
    float rate;
    int DD;public:
    Share () {Code = 1000; Rate = 100; DD = 1;}
    void GetCode (long int C, float R)
    {
        Code = C;
        Rate = R;
        {
            void Update (int Change, int D)
            {
                Rate += Change;
                DD = D;
            }
        }
        void Status ()
        {
            cout<< "Date:"<<DD<<endl;
            cout<<Code<< "#"<<Rate<<endl;
        }
    };
    void main ()
    {
        Share S, T, U;
```

```
S.GetCode (1324, 350);
T.GetCode (1435, 250);
S.Update (50, 28);
U.Update (-25, 26);
S.Status ();
T.Status ();
U.Status ();
}
```

Ans: Date:28

1324#400

Date: 1

1435#250

Date: 26

1000#75

(f) Look at the following C++ code and find the possible output(s) from the options (i) to (iv) following it. Also, write the maximum and the minimum values that can be assigned to the variable PICKER. 2 Marks

Note: Assume all the required header files are already being included in the code.

The function random (n) generates an integer between 0 and n-1.

```
void main ()
```

```
{
```

```
randomize ();
```

```
int PICKER;
```

```
PICKER = 1+ random (3);
```

```
char COLOR [][5] = {"BLUE", "PINK", "GREEN", "RED"};
```

```
{
```

```
for (int J = 0; J<=I; J++)
```

```
Cout<<COLOR [J];
```

```
cout<<endl;
```

}
}

(i)	(ii)	(iii)	(iv)
PINK	BLUE	GREEN	BLUE
PINK-GREEN	BLUEPINK	GREEN-RED	BLUE PINK
PINK-GREEN-RED	BLUE PINK GREEN BLUE PINK GREEN RED		BLUE PINK GREEN

Ans: (iv) BLUE

BLUE PINK

BLUE PINK GREEN

2. (a) Write any four important characteristics of Object Oriented Programming? Give example of any one of the characteristics using C++. 2 Marks

Ans: Four characteristics of object oriented Programming are:

- (a) Encapsulation
- (b) Abstraction
- (c) Inheritance
- (d) Polymorphism

Examples:

Class Student

```
{
    int Roll;
    char name[20];
    public:
    void input( )
    {
```

```

        cout<<"Name:"<<name<<endl;
        cout<<"Roll:"<<roll<<endl;
    }
}
Void main( )
{
    Students s;
    s.input( );
    s.display( );
}

```

In the above program, by declaring the class we are using the concept of encapsulation and in the main method while calling the functions using the object of the class is abstraction.

(b) Observe the following C++ code and answer the questions (i) and (ii). Assume all necessary files are included:

```

class Book
{
    long Code;
    char Title [20];
    float Price;
public:
    Book ()           //Member Function 1
    {
        cout<< "Bought"<<endl;
        Code = 10; strcpy (title, "No Title"); Price = 100;
    }
    BOOK (int C, char T [], float P) //Member Function
    {
        Code = C;
        strcpy (Title, T);
    }
}

```

```

Price = P;
}

void Update (float p) //Member Function 3
{
Price +=P;
}

void Display () //Member Function 4
{
cout<<Code<< “:”<<Title<< “:”<<Price<<endl;
}

~Book () //Member Function 5
{
cout<< “Book Discarded!”<<endl;
}
};

void main () //Line 1
{ // Line 2
Book B, C (101, “Truth”, 350); // Line 3
for (int I= ); I<4; I++) // Line 4
{ // Line 5
B.Update (50); C.Update (20); // Line 6
B.Display (); C.Display (); // Line 7
} // Line 8
} // Line 9

```

(i) Which specific concept of object oriented programming out of the following illustrated by Member Function 1 and Member Function 2 combined together? 1 Mark

- **DataEncapsulation**
- **Polymorphism**

- Inheritance
- Data Hiding

Ans: Polymorphism

(ii) How many times the message “Book Discarded!” will be displayed after executing the above C++ code? Out of line 1 to Line 9, which line is responsible to display the message “Book Discarded!” **1 Mark**

Ans: 2 times, once for each object when the destructor will be invoked.

(c) Write the definition of a class CITY in C++ with following description: **4 Marks**

Private Members

- Ccode //Data member for City Code (an Integer)
- CName // Data member for City Name (a string)
- Pop // Data member for Population (a long int)
- KM // Data member for Area Coverage (a float)
- Density // Data member for Population Density (a float)
- DenCal () // A member function to calculate
// Density as Pop KM

Public Members

- Record () // A function to allow user to enter values of
// Acode, Name, Pop, KM and call DenCal () function
- View () // A function to display all the data members
// also display a message “Highly Populated City”
// if the Density is more than 10000

Ans: class CITY

```
{
    Private:
    int CCode;
    char CName[20];
    long pop;
    float KM;
```

```
float Density;
DenCal()
{
Density=pop/KM
}
public:
Record()
{
    cout<<"Enter Code, Name,Pop,Km:"<<endl;
    cin>>CCode>>CName>>pop>>KM;
    DenCal();
}
View()
{
    cout<<"CCode"<<"\t"<<CName<<"\t"<<pop<<
    "\t"<<KM<<endl;
    cout<<CCode<<"\t"<<CName<<"\t"<<pop<<"\t
    "<<KM<<endl;
    if(Density > 1000)
    cout<<"Highly Populated"<<endl;
}
}
```

(d) Answer the questions (i) to (iv) based on the following:

4 Marks

class ITEM

{

int Id;

char IName[20];

protected:

float Qty;

```
public:
ITEM ();
void Enter (); void View ();
};
class TRADER
{
int DCode;
protected:
char Manager [20];
public:
TRADER ();
void Enter ();
void View ();
};
class SALEPOINT : public ITEM, private TRADER
{
char Name [20], Location [20];
public:
SALEPOINT ();
void EnterAll ();
void ViewAll ();
};
```

(i) Which type of Inheritance out of the following is illustrated in the above example?

- Single Level Inheritance**
- Multi Level Inheritance**
- Multiple Inheritance**

Ans: Multiple Inheritance

(ii) Write the names of all the data members, which are directly accessible from the member functions of class SALEPOINT.

Ans: Id, IName, Qty, Name, Location

(iii) Write the names of all the member functions, which are directly accessible by an object of class SALEPOINT.

Ans: ITEM::Enter, ITEM::View, EnterAll, ViewAll

(iv) What will be order of execution of the constructors, when an objects of class SALEPOINT is declared?

Ans: SALEPOINT()
ITEM()

3. (a) Write the definition of a function Fix Salary (float Salary [], int N) in C++, which should modify each element of the array Salary having N elements, as per the following in the memory along the row with each of its element rules:

2 Marks

Existing Salary Values	Required Modification in Value
If less than 1,00,000	Add 35% in the existing value
If >=1,00,000 and <20,000	Add 30% in the existing value
If >=2,00,000	Add 20% in the existing value

Ans: void FixSalary(float Salary[], intN)

```
{
    float d;
    for(int i=0;i<N;i++)
    {
        If(Salary[i]<1000000)
            d=35.0/100;
        else if(Salary[i]<2000000)
            d=30.0/100;
        else
```

```

        d=20.0/100;
        Salary[i]+=d;
    }
}

```

(b) R [10] [50] is a two dimensional array, which is stored occupying 8 bytes, find the address of the element R [5] [15], if the element R [8][10] is stored at the memory location 45,000. 3 Marks

Ans: address= $45000 + 8 \times (5-8) \times 50 + (15-10)$
 $= 45000 + 8 \times (-150 + 5)$
 $= 45000 + 8 \times (-145)$
 $= 45000 - 1160 = 43840$

(c) Write the definition of a member function DELETE () for a class QUEUE in C++, to remove a product from a dynamically allocated Queue of products considering the following code is already written as a part of the program. 4 Marks

```

struct PRODUCT
{
    int PID; char PNAME [20];
    PRODUCT *Next;
};
class QUEUE
{
    PRODUCT *R, *F;
public:
    QUEUE () {R= NULL; F = NULL;}
    void INSERT ();
    void DELETE ();
    ~ QUEUE ();
};

```

Ans: void DELETE()

```
{
    If(F==R)
    {
        Cout<<"Empty queue";
        return;
    }
    F=F.next;
}
```

(d) Write definition for a function DISPMID (int A [] [5], int R, int C) in C++ to display the elements of middle row and middle column from a two dimensional array A having R number of rows and C number of columns. For example, if the content of array is as follows: **3 Marks**

215	912	516	401	515
103	901	921	802	601
285	209	609	360	172

The function should display the following as output: 103 901 921 802 601 ‘
516 921 609

Ans: void DISPMID (int A [] [5], int R,int C)

```
{
    for (int i=0;i<c;i++)
        cout<<A[R/2][i]<<" ";
    cout<<endl;
}
```

(e) Convert the following Infix expression to its equivalent Post fix expression, showing the stack contents for each step of conversion. **2 Marks**

P/ (Q-R)*S+T

Ans: P/(Q-R)*S+T

= P/(QR-)*S+T

= (PQR-/*)*S+T

= PQR-/S*+T

= PQR-/S*T+

4. (a) Write a function definition of DISP3CHAR() in C++ to read the content of a text file KIDINME.TXT, and display all those words, which has three characters in it. Example: If the content of the file KIDINME.TXT is as follows: 2 Marks

When I was a small child, I used to play in the garden with my grand mom. Those days were amazingly funful and I remember all the moments of that time.

The following DISP3CHAR () should display the following:

was the mom and all the

Ans: void DISP3CHAR ()

```
{
    ifstream in;
    char txt[100];
    int a, i, j, k;
    char word[3],c;
    in.open ("KIDINMETXT",ios::in);
    if(!in)
    {
        cout<<"Error in opening the file";
        exit (1);
    }
    While(1)
    {
        in.getline (txt, 100);
        for (i=0;i<100;)
```

```

{
    c=txt[i];
    if (c==' ')
    {
        for (j=i;txt[j]!=' ';j++);
        if (j - i==4)
        {
            for(k=i+1,a=0;k<j;k++)
            word[a]=txt[k];
            word[a] = '\0';
            cout<<word<<" ";
        }
        i=j;
    }
}
}

```

(b) Write a definition for function ONOFFER() in C++ to read each object of a binary file TOYS.DAT, find and display details of those toys, which has status as "ON OFFER". Assume that the file TOYS.DAT is created with the help of objects of class, TOYS, which is defined below: 3 Marks

class TOYS

```

{
int TID; char Toy [20], Status [20]; float MRP;
public:
void Getinstock ()
{
cin>>TID; gets (Toy); gets (Status); cin>>MRP;
}
void view ()

```

```
{
cout<<TID<< ":"<<Toy<< ":"<<MRP<< ":"<<Status<<endl;
}
char *Seeoffer () {return Status;}
};
```

Ans: void NOOFFER()

```
{
    ifstream in (" TOYS.DAT",ios::binary);
    TOYS obj;
    While(in)
    {
        in.read((char*)&obj,sizeof(obj));
        if(strcmp(obj.Status,"ON OFFER"))
        cout<<obj.TID<<" "<<obj.Toy<<" "<<obj
            Status<<" "<<obj.MRP<<endl;
    }
    In.close();
}
```

(c) Find the output of the following C++ code considering that the binary file CLIENT. DAT exists on the hard disk with a data of 1000 clients. 1 Mark

class CLIENT

```
{
int Ccode; char CName [20];
public:
void Register (); void Display ();
};
void main ()
{
fstream CFile;
```

```
CFile.open ("CLIENT.DAT", ios: : binary/ios: :in);
```

```
CLIENT C;
```

```
CFile.read ((char*) & C, sizeof (c));
```

```
cour<< "Rec:"<<CFile.tellg ()/sizeof (c) <<endl;
```

```
CFile.read ((char*) & C, sizeof (C));
```

Ans: Rec: 2

Rec: 4

SECTION - B

1. (a) Out of the following, find those identifiers, which cannot be used for naming Variables or Functions in a Python program: 2 Marks

Total*Tax, While, class, switch, 3rd Row, finally, Column 31, _Total

Ans: Total*Tax While, class, switch 3rdRow, finally

(b) Name the Python Library modules which need to be imported to invoke the following functions:

(i) sqrt ()

(ii) dump ()

Ans: (i) sqrt() → math module

(ii)dump() → pickle module

(c) Rewrite the following code in Python after removing all syntax error(s). Underline each correction done in the code. 2 Marks

for Name in [Ramesh, Suraj, Priya] IF Name [0] = 'S' : Print (Name)

Ans: for Name in ["Ramesh", "Suraj", "Priya"]:// "can be used if Name [0] == 'S': print (Name)

(d) Find and write the output of the following Python code: 2 Marks

```
Values = [10, 20, 30, 40]
```

```
for Val in Values:
```

```
    for I in ranges(1,val%9);
```

```
        print (I,"*", ends="")
```

```
    print ( )
```

Ans:

	0	
	(1, *)	0
1*	0	(1, *)
1*	(1, *)	(1*2*3)
2*	(2, *)	(1*2*3*)
1*	0	
2*	(1, *)	1*
3*	(2, *)	1*2*
	(3, *)	1*2*3*
	0	

(e) Find and write the output of the following Python code:

3 Marks

Class Book

def __init__(self, N=100,S="Python"):#contructor

self.Bno=N

self.BName=S

def Assign(self, N,S):

self.Bno=self.bNo+N

self.BName=S+self.BName

def Showval(self):

print (self.Bno,"#",self.BName)

s=Book ()

t=Book (200)

u=Book (300, "Made Easy")

s.ShowVal ()

t.ShowVal ()

u.ShowVal ()

Ans:

Python 2.7 Output	Other Versions output
100 #Python	(100, '#', 'Python')
100 #Python	(200, '#', 'Python')
300 #Made Easy	(300, '#', 'Made Easy')

105 #Made Python	(105, '#', 'Made Python')
215 #Easy Python	(215, '#', 'Easy Python')
325 #Made EasyMade Easy	(325, '#', 'Made' EasyMade Easy')

(f) What are the possible Outcomes, (s) executed from the following code? Also specify the maximum and minimum values that can be assigned to variable PICKER. **2 Marks**

```

import random
PICKER=random.randint(0,3)
COLOR= ["BLUE","PINK","GREEN","RED"];
for I in COLOR:
    for J in range ( 1, PICKER):
        print (I, end="")
    print ( )

```

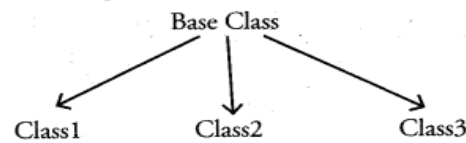
(i)	(ii)	(iii)	(iv)
BLUE	BLUE	PINK	BLUEBLUE
PINK	BLUEPINK	PINKGREEN	PINKPINK
GREEN	BLUEPINKGREEN	GREENRED	GREENGREEN
RED	BLUEPINKGREENRED		REDRED

Ans: Option (i) and (iv) are possible

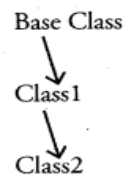
2. (a) What is the difference between Multilevel and Multiple inheritance? Give suitable examples to illustrate both. **2 Marks**

Ans: In Multiple Inheritance we can inherit more than one class in the same class where as in Multi-Level Inheritance one class can inherit only one base class and the derived class can become base class of some other class same.

In Multiple inheritance structure is like this :



In Multi Level Inheritance structure is like this:



(b)What will be the output of the following Python code considering the following set of inputs? 2 Marks

AMAR

THREE

A123

1200

Also, explain the try and except used in the code.

Start = 0

while True:

try:

Number = int (raw_input ("Enter Number"))

break

except ValueError:

Start = Start + 2

print ("Re-enter an integer")

print (Start)

Ans: Enter a number: AMAR

Re-enter an integer 2

Enter a number: THREE

Re-enter an integer 4

Enter a number: A123

Enter a number:1200

Try part is used to execute the statement and try to read an integer value and except part is executing every time when the program is reading every time when the program is reading a non-integer value and redirecting to the beginning of the loop.

(c) Write a Class CITY in Python with following specifications. 4 Marks

Instance Attributes

- **Code** **# Numeric value**
- **CName** **# String Value**
- **Pop** **# Numeric Value for Population**
- **KM** **# Numeric Value**
- **Density** **# Numeric value for Population Density**

Methods:

- **DenCal ()** **# Method to Calculate Density as Pop/KM**
- **Record ()** **# Method to allow user to enter values Ccode, CName, Pop, KM and call DenCal () method**
- **View ()** **# Method to display all the members also displays a message “Highly Populated City” if the Density is more than 10000.**

Ans: class CITY:

```
def __init__(self, Ccode,CName,Pop,KM,Density)
```

```
def DenCal()
```

```
self.Density=self.Pop/self.KM
```

```
def Record (self,C,N,P,K)
```

```
self.Ccode=C
```

```
self.CName=N
```

```
self.Pop=p
```

```
self.KM=k
```

```
def View()
```

```
print (CCode,"#",CName,'#',Pop,'#'KM,'#',Density)
```

```
if self.Density>10000
```

```
print (“Highly Populated City”)
```

(d) How do we implement abstract method in Python? Give an example for the same. 2 Marks

Ans: Abstract method in Python

```

import abc

class Shape (object):
    __metaclass__=abc.ABCMeta
    @abc.abstractmethod
    def method_to_implement(self,input);
        """Method documentation"""
        return;

```

(e) what is the significance of super() method? Give an example for the same. 2 Marks

Ans: Super method in Python

```

class Base (object):
    def __init__ (self):
        print "Base created"
class ChildA(Base):
    def __init__ (self):
        Base.__init__ (self)
class ChildB (Base):
    def __init__ (self):
        super (ChildB,self).__init__()
ChildA ()
ChildB ()

```

3. (a) What will be the status of the following list after the 'First, Second and Third pass of the selection sort method used for arranging the following elements in descending order? 3 Marks

Note: Show the status of all the elements after each pass very clearly underlining the changes. 12, 14, -54, 64, 90, 24

Ans:

	12	14	-54	64	90	24
Pass 1	90	14	-54	64	12	24
Pass 2	90	64	-54	14	12	24
Pass 3	90	64	24	14	12	-54

(b) For a given list of values in descending order, write a L method in Python to search for a value with the help of Binary search method. The method should return position of the value and should return —1 if the value; not present in the list. 2 Marks

Ans: def binarysrch (nums, x):

high = len (nums)

low = 0

while low<high:

mid = (low + high) //2

midval = nums[mid]

if midval>x;

low = mid + 1

elif midval <x:

high = mid

else:

return mid

return -1

(c) Write Insert (City) and Delete (City) methods in Python to add City and Remove City considering them to act as Insert and Delete operations of the data structure Queue. 4 Marks

Ans: class queue:

city = [] def Insert (self):

a = raw_input ("Enter city")

queue.city.append (a)

def Delete (self):

if (queue.city == []):

```
print "Queue empty"
else:
print "Deleted element is", queue.city[0]
queue.city.delete ()
```

(d) Write a method in Python to find and display the prime numbers between 2 to N. Pass N as argument to the method. 3 Marks

Ans: def prime (N):
for a in range (2, N):
Prime = 1
for I in range (2,a):
if a%i ==0:
Prime = 0
if Prime ==1:
print a

(e) Evaluate the following postfix notation of expression. Show status of stack after every operation. 2 Marks

12, 2, /, 34, 20, -, +, 5, +

Ans:

Element	Stack
12	12
2	12, 2
/	6
34	6, 34
20	6, 34, 20
-	6, 14
+	20
5	20, 5
+	25

Final Result = 25

4. (a) Write a statement in Python to perform the following operations: 1 Mark

- To open a text file "MYPET.TXT" in write mode

- To open a text file “MYPET.TXT” in read mode

Ans:

- f1=open ('MYPET.TXT','W')
- f2=open ('MYPET.TXT','r')

(b) Write a method in Python to write multiple line of text contents into a text file daynote.txt line. 2 Marks

Ans: def.write (): f = open (“daynote.txt”, ‘w’)

while True;

line = raw_input (“Enter line”)

f.write (line)

choice = raw_input(“Are there more lines”)

if choice == ‘N’:

break

f.close ()

(c) Consider the following definition of class Employee, write a method in Python to search and display the content in a pickled file emp.dat, where Empno is matching with ‘A0005’. 3 Marks

Class Employee:

def_init_(self,E,NM):

self.Empno=E

def display(self):

print(self.Empno,” – “, self.ENAME)

Ans: def.search ();

f= open (“emp.dat”, ‘rb’)

try:

while True:

e = pickle.load (f)

If e.Empno == ‘A0005’;

e.display ()

except EOFError;

pass

f.close ()

SECTION – C

5. (a) Observe the following PARTICIPANTS and EVENTS' tables carefully and write the name of the RDBMS operations which will be used to produce the output as shown in RESULT? Also, find the Degree and Cardinality of the result. 2 Marks

PARTICIPANTS

PNO	NAME
1	Aruanabha Tariban
2	John Fedricks
3	Kanti Desai

EVENTS

EVENTCODE	EVENTNAME
1001	IT Quiz
1002	Group Debate

RESULT

PNO	NAME	EVENT CODE	EVENT Name
1	Aruanabha Tariban	1001	IT Quiz
1	Arunabha Tariban	1002	Group Debate
2	John Fedricks	1001	IT Quiz
2	John Fedricks	1002	Group Debate
3	Kanti Desai	1001	IT Quiz
3	Kanti Desai	1002	Group Debate

Ans: PRODUCT

Degree=4

Cardinality=6

(b) Write SQL queries for (i) to (iv) and find outputs for SQL queries (v) to (viii), which are based on the tables. 6 Marks

VCODE	VEHICLE ETYPE	PERKM
V01	VOLVO BUS	150
V02	AC DELUXE BUS	125
V03	ORDINARY BUS	80
V05	SUV	30

V04	CAR	18
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Note: PERKM is Freight Charges per kilometer

Table: TRAVEL

CNO	CNAME	Travel date	KM	VCODE	NOP
101	K.Niwal	2015-12-13	200	V01	32
103	Fredrick Sym	2016-03-21	120	V03	45
105	Hitesh Jain	2016-04-03	450	V02	42
102	Ravi Anish	2016-01-13	80	V02	40
107	John Malina	2015-02-10	65	V04	2
104	Sahanubhuti	2016-01-28	90	V05	4
106	Ramesh Jaya	2016-04-06	100	V01	25

NOTE:

- KM is kilometers travelled
- NOP is number of passengers travelled in vehicle

(i) To display CNO, CNAME, TRAVELDATE from the table TRAVEL in descending order of CNO.

Ans: SELECT CNO, CNAME, TRAVELDATE FROM TRAVEL ORDERBY CNO DESC

(ii) To display the CNAME of all the customers from the table TRAVEL who are travelling by vehicle with code V01 or V02.

Ans: SELECT * FROM TRAVEL WHERE KM>120 ORDERBY NOP

(iii) To display the CNO and CNAME of those customers from the table TRAVEL who travelled between '2015-12-31' and '2015-05-01'.

Ans: SELECT CNO, CNAME FROM TRAVEL WHERE TRAVELDATE BETWEEN '2015-05-01' AND '2015-12-31'

(iv) To display all the details from the table TRAVEL for the customers, who have travel distance more than 120 KM in ascending order of NOP.

Ans: SELECT * FROM TRAVEL WHERE KM>120 ORDERBY NOP

(v) Select Count (*), Vcode From Travel Group By Vcode Having Count (*)>1;

Ans: V01 2

V02 2

(vi) Select Distinct Vcode From Travel;

Ans: V01

V03

V02

V04

V05

(vii) select a.vcode, cname, vehicletype from travel a, vehicle b where a.vcode=b.vcode and km<90;

Ans: V02 Ravi Anish AC DELUXE BUS

V04 John Malina CAR

(viii) Select Cname, Km* Perkm From Travel A, Vehicle B Where A. Vcode=B.Vcode And A. Vcode = 'V05';

Ans: Sahanubhuti 2700

6. (a) Verify the following using Boolean Laws.

2 Marks

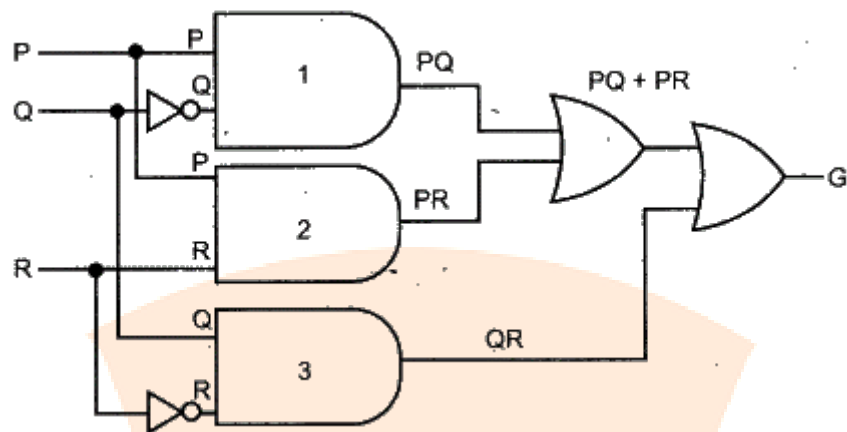
$$X' + YZ = X'.Y'.Z' + X'.Y.Z' + X'.Y.Z + X'.Y'.Z + X.Y'.Z$$

Ans: $X' + Y'.Z = X'.Y.Z + X'.Y.Z' + X'.Y'.Z + X.Y'.Z$

$$\begin{aligned} \text{RHS} &= X'.Y.Z + X'.Y.Z' + X'.Y.Z + X.Y'.Z \\ &= X' (Y'.Z' + Y.Z' + Y.Z + Y'.Z) + X.Y'.Z \\ &= X' (Y' (Z' + Z) + Y (Z' + Z) + X.Y'.Z) \\ &= X' (Y' + Y) + X.Y' + X.Y'.Z \\ &= X' + X.Y'.Z \\ &= X' + Y'.Z = \text{L.H.S. (hence Proved)} \end{aligned}$$

(b) Write the Boolean Expression for the result of the Logic Circuit as shown below:

2 Marks



Ans: $G = P.Q' + P.R + Q.R$

(c) Derive a Canonical SOP expression for a Boolean function G, represented by the following truth table: 1 Mark

A	B	C	G (A,B,C)
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

Ans: $G = A'.B'.C' + A'.B.C' + A.B.C' + A.B.C$

(d) Reduce the following Boolean Expression to its simplest form using K-Map: 3 Marks

$$F(P,Q,R,S) = \Sigma (0,4,5,8,9,10,11,12,13,15)$$

Ans: $P'Q' \quad P'Q \quad PQ \quad PQ'$

R'S'	1	1	1	1
	0	4	12	8
R'S		1	1	1
RS	1	5	13	9
RS'			1	1
	3	7	15	11
				1

2	6	14	10
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Quad 1(0, 4, 12, 8) = $R'.S'$

Quad 2(4, 5, 12, 13) = $Q.R'$

Quad 3(13, 15, 9, 11) = $P.S$

Quad 4(8, 9, 11, 10) = $P.Q'$

Reduced expression $F(P, Q, R, S) = R'.S' + Q.R' + P.S + P.Q'$

7. (a) Differentiate between PAN and LAN types of networks. 1 Mark

Ans: PAN (Personal Area Network) is a small network that connects a desktop to a laptop, phone, tablet, or other device for personal use, whereas LAN (Local Area Network) is a larger network that spans a building and connects all computers and peripherals through cables.

(b) Which protocol helps us to transfer files to and from a remote computer? 1 Mark

Ans: FTP – File Transfer Protocol

(c) Write two advantages of 3G over 2G Mobile Telecommunication Technologies in terms of speed and services? 1 Mark

Ans: Advantages of 3G over 2G Mobile Telecommunications are:

- (i) The network is quite fast.
- (ii) Wireless broadband will be available to customers.
- (iii) Customers can watch programmes through video or satellite.
- (iv) All the facilities can be used at same time.

(d) Write two characteristics of Web 2.0.

Ans: Two characteristics of Web 2.0 are:

(i) Folksonomy: Traditional web sites such as Yahoo and DMOZ use pre-defined classifications of information such as category and sub category, however Web 2.0, by not adhering to the established classification structure, allows users to construct their own free arrangements of information.

(ii) Rich User experience: Traditional web built with HTML and CSS, CGI and had been offered as a static page on the other hand Web 2.0 uses AJAX (Asynchronous JavaScript + XML) presenting dynamic and rich user experience to the users.

(e) What is the basic difference between Computer Worm and Trojan Horse?
1 Mark

Ans: Basic difference between Computer Worm and Trojan Horse is there in their functionality. Worm is a kind of program that replicates itself from one system to another without any aid of the host file whereas Trojan Horse is a program that claims to be something useful but actually harmful to the system and also transfers the system's secret to the sender.

(f) Categorise the following under Client side and Server Side script category?
1 Mark

(i) Java Script

Ans: Server side script

(ii) ASP

Ans: Client side script

(iii) VB Script

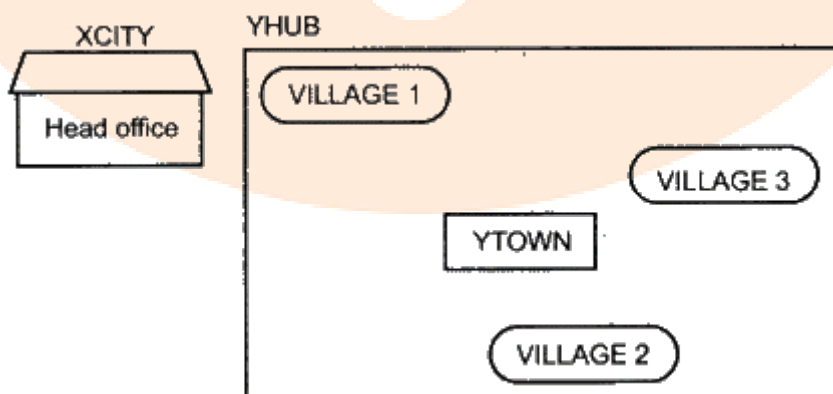
Ans: Server side script

(iv) JSP

Ans: Client side script

(g) Intelligent Hub India is a knowledge Community aimed to uplift the standard of skills and knowledge in the society. It is planning to set up its training centres in multiple towns and villages pan India with its head offices in the nearest cities. They have created a model of their network with a city, a town and 3 villages as follows.

As a network consultant, you have to suggest the best network related solutions for their issues/problems raised in (i) to (iv), keeping in mind the distances between various locations and other given parameters.



Shortest distances between various locations:

VILLAGE 1 to YTOWN	2 KM
VILLAGE 2 to YTOWN	1.5 KM
VILLAGE 3 to YTOWN	3 KM
VILLAGE 1 to VILLAGE 2	3.5 KM
VILLAGE 1 to VILLAGE 3	4.5 KM
VILLAGE 2 to VILLAGE 3	3.5 KM
CITY Head Office to YHUB	30 KM

Number of Computers installed at various locations are as follows:

YTOWN	100
VILLAGE 1	10
VILLAGE 2	15
VILLAGE 3	15
CITY OFFICE	5

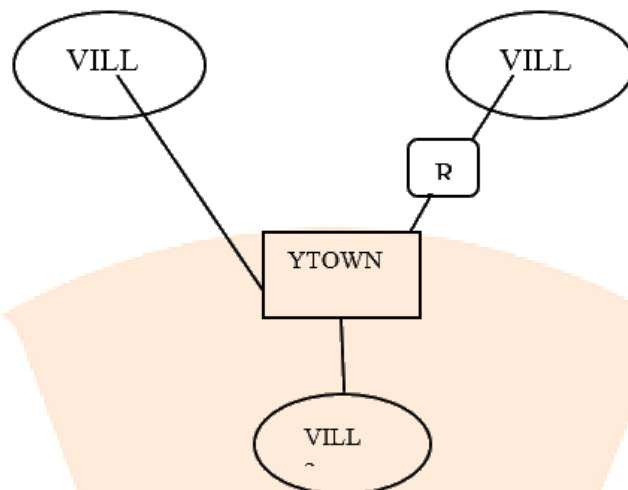
Note: In villages, there are community centres, in which one room has been given training center to this organization to install computers. The organization has got financial support from the government and top IT companies.

(i) Suggest the most appropriate location of the SERVER in the YHUB (out of the 4 locations), to get the best and effective connectivity. Justify your answer. **1 Mark**

Ans: Server must be placed at YTOWN as maximum computers are there thus decreasing the cabling cost and increasing the efficiency of the computers in the network.

(ii) Suggest the best wired medium and draw the cable layout (location to location) to efficiently connect various locations with the YHUB. **1 Mark**

Ans: The following structure would be the most economical way to connect all the villages with YTOWN with reasonable high speed fibre optic cables for VILL1 and VILL2 and radio wave transmission for VILL3 to connect to YTOWN.



(iii) Which hardware device will you suggest to connect all the computers within each location of YHUB? 1 Mark

Ans: Switch would be needed in all the buildings as more than 10 computers are there in the building.

(iv) Which service/protocol will be most helpful to conduct live interactions of Experts from Head office and people at YHUB locations? 1 Mark

Ans: Video conferencing would be the best option.